Torsten Scholak, Thomas Wellens, and Andreas Buchleitner:

Exciton transport in ordered and disordered samples of cold Rydberg atoms

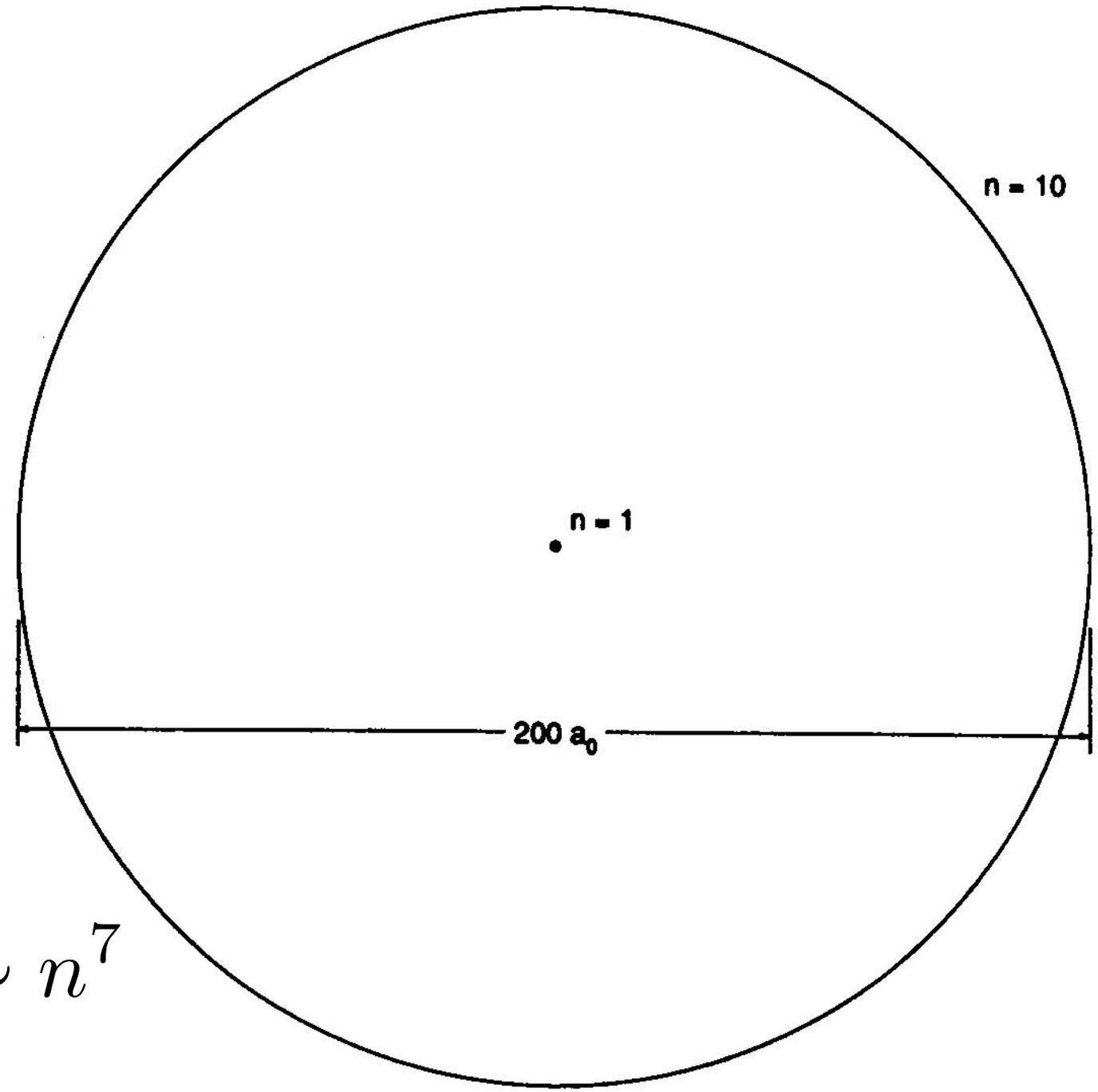
QOS
Quantum Optics and Statistics
Albert-Ludwigs-Universität Freiburg

outline

- Rydberg gases
- Hamiltonian
- results

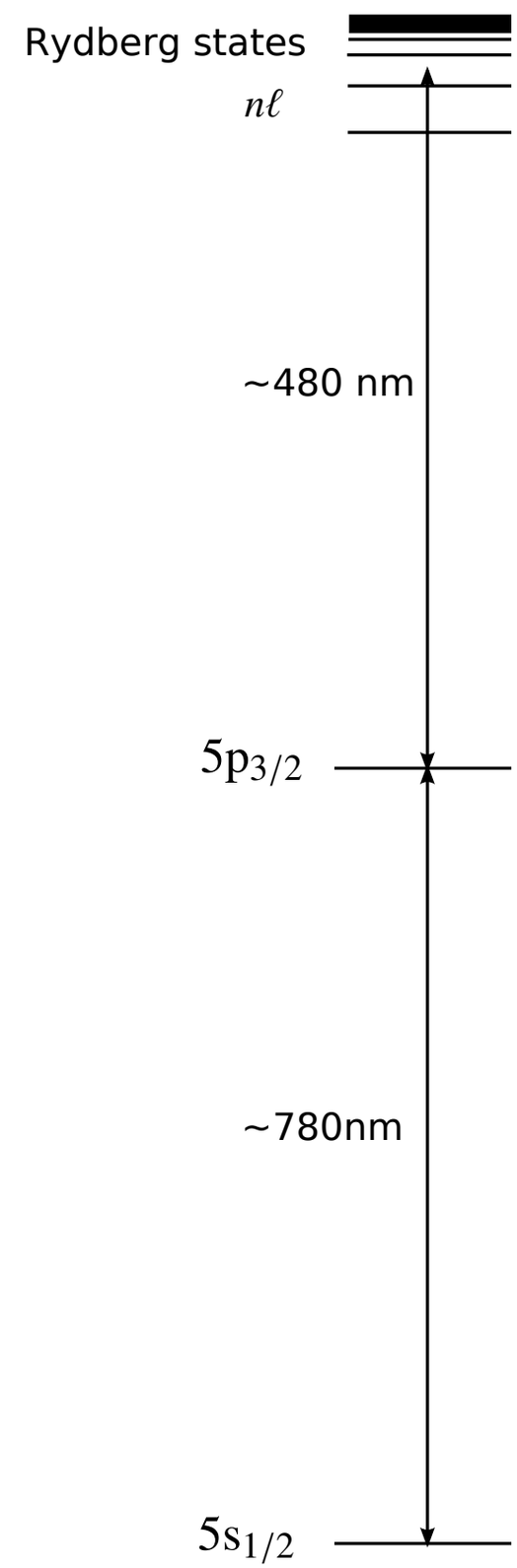
Rydberg atoms

- highly-excited electronic states, radius $\sim n^2$
- long lifetime $\sim n^3$
- large polarizability $\sim n^7$



T. F. Gallagher, *Rydberg Atoms* (Cambridge University Press, Cambridge, 1994)

frozen gases



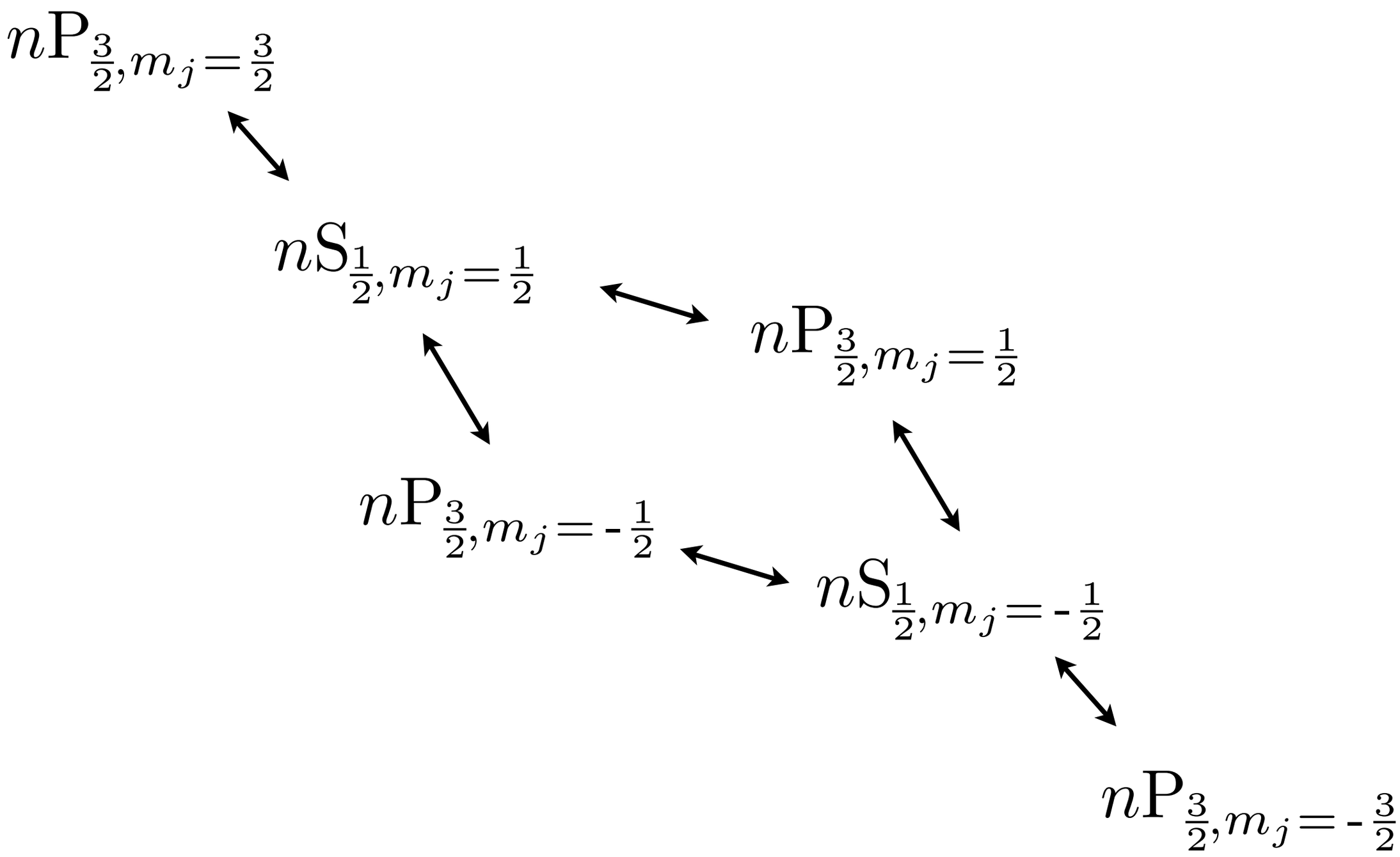
- magneto-optical trap, $100\mu\mathrm{K}$
- densities $> 10^7\mathrm{cm}^{-3}$
- excite to Rydberg states
- disordered crystal
- long-range interactions

T. Amthor, Ph.D. thesis, Albert-Ludwigs-Universität Freiburg, (2008)

interactions

$$\hat{V}_{ij} = \sum_{i \neq j} \left(\frac{\hat{\boldsymbol{\mu}}_i \cdot \hat{\boldsymbol{\mu}}_j}{R_{ij}^3} - 3 \, \frac{\left(\hat{\boldsymbol{\mu}}_i \cdot \boldsymbol{R}_{ij}\right)\left(\boldsymbol{R}_{ij} \cdot \hat{\boldsymbol{\mu}}_j\right)}{R_{ij}^5} \right)$$

interactions



interactions

$$\mathrm{P}_+ \leftrightarrow \mathrm{S}_+$$

$${}_{ij}\langle \mathrm{S}_+\mathrm{P}_+|\hat{V}|\mathrm{P}_+\mathrm{S}_+\rangle_{ij} \propto \frac{\cos^2\vartheta_{ij} - \frac{1}{3}}{R_{ij}^3}$$

$${}_{ij}\langle \mathrm{S}_\pm\mathrm{P}_\mp|\hat{V}|\mathrm{P}_\pm\mathrm{S}_\mp\rangle_{ij} \propto \frac{\sin^2\vartheta_{ij}\,\mathrm{e}^{-2i\varphi_{ij}}}{R_{ij}^3}$$

$${}_{ij}\langle \mathrm{S}_-\mathrm{P}_-|\hat{V}|\mathrm{P}_-\mathrm{S}_-\rangle_{ij} \propto \frac{\cos^2\vartheta_{ij} - \frac{1}{3}}{R_{ij}^3}$$

$$\mathrm{S}_- \leftrightarrow \mathrm{P}_-$$

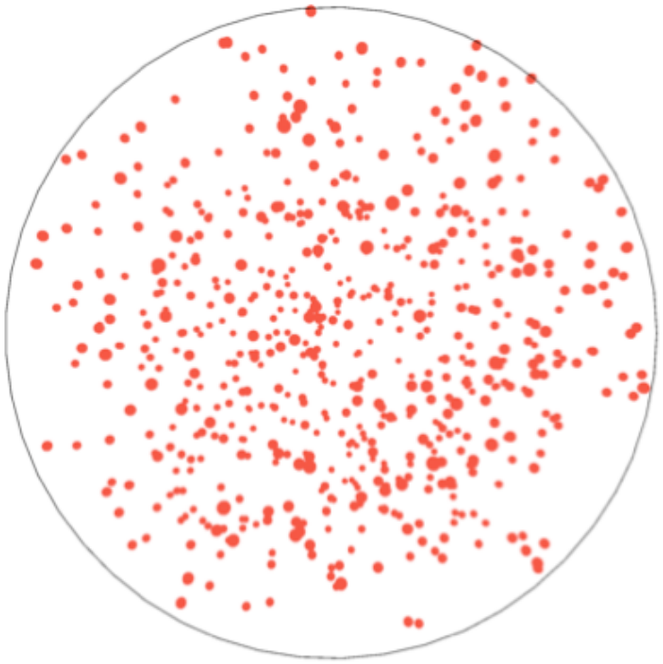
Hamiltonian 1-2-3

- randomly predefine atomic species
- just one P excitation

$$|\mathrm{P}_+\mathrm{S}_-\mathrm{S}_-\ldots\mathrm{S}_+\ldots\mathrm{S}_+\rangle := |1\rangle$$

$$|\mathrm{S}_+\mathrm{P}_-\mathrm{S}_-\ldots\mathrm{S}_+\ldots\mathrm{S}_+\rangle := |2\rangle$$

$$|\mathrm{S}_+\mathrm{S}_-\mathrm{S}_-\ldots\mathrm{P}_+\ldots\mathrm{S}_+\rangle := |i\rangle$$



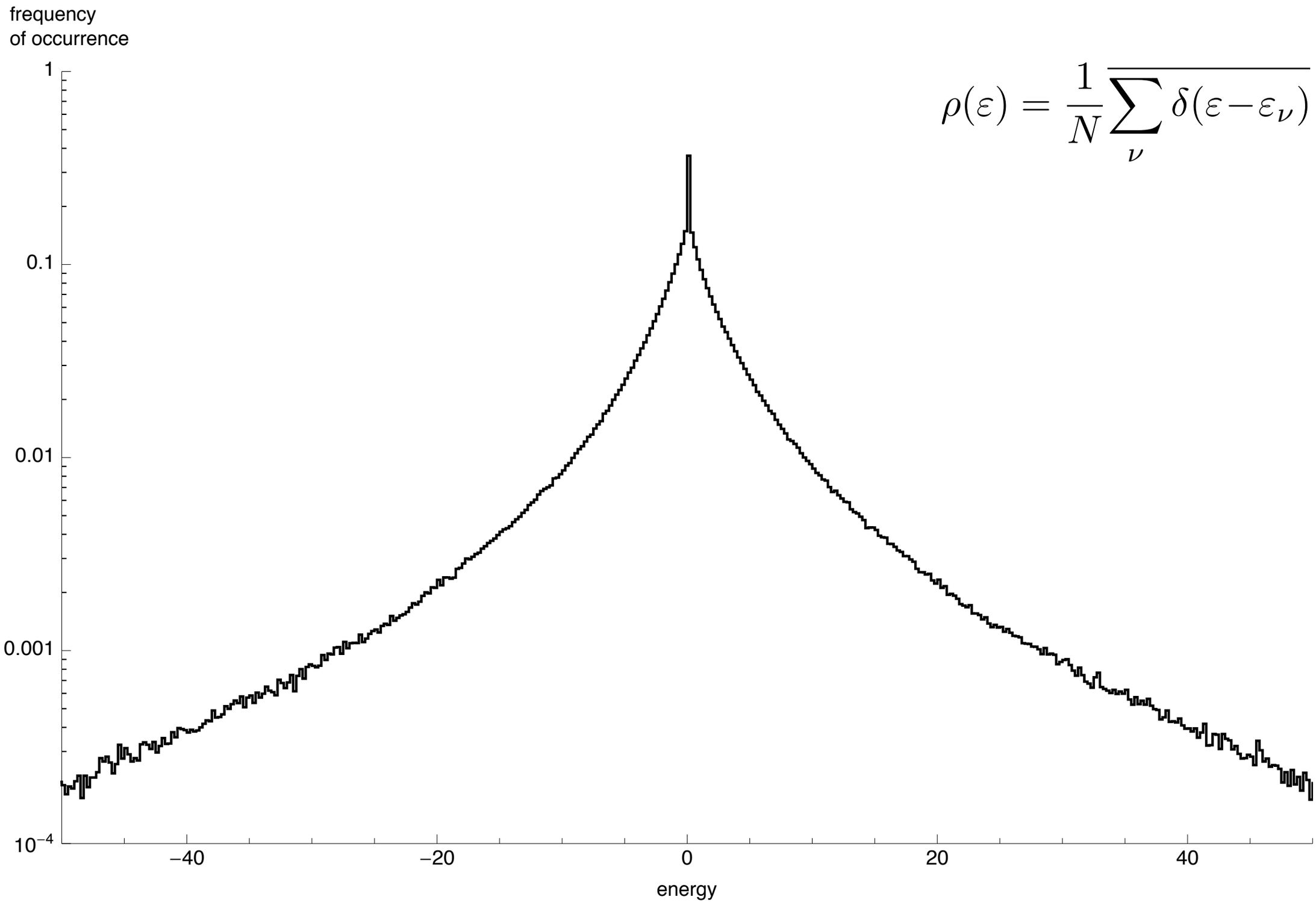
- arrive at tight-binding Hamiltonian

$$H_{ij} = \langle i|\hat{H}|j\rangle = (\varepsilon_{\mathrm{P}} - \varepsilon_{\mathrm{S}})\delta_{ij} + (1-\delta_{ij})V_{ij}$$

- perform ensemble average

density of states

$$\rho(\varepsilon) = \frac{1}{N}\overline{\sum_{\nu} \delta(\varepsilon - \varepsilon_\nu)}$$



level spacings

- Poisson

$$p(s) = \mathrm{e}^{-s}$$

- Gaussian Orthogonal (GOE)

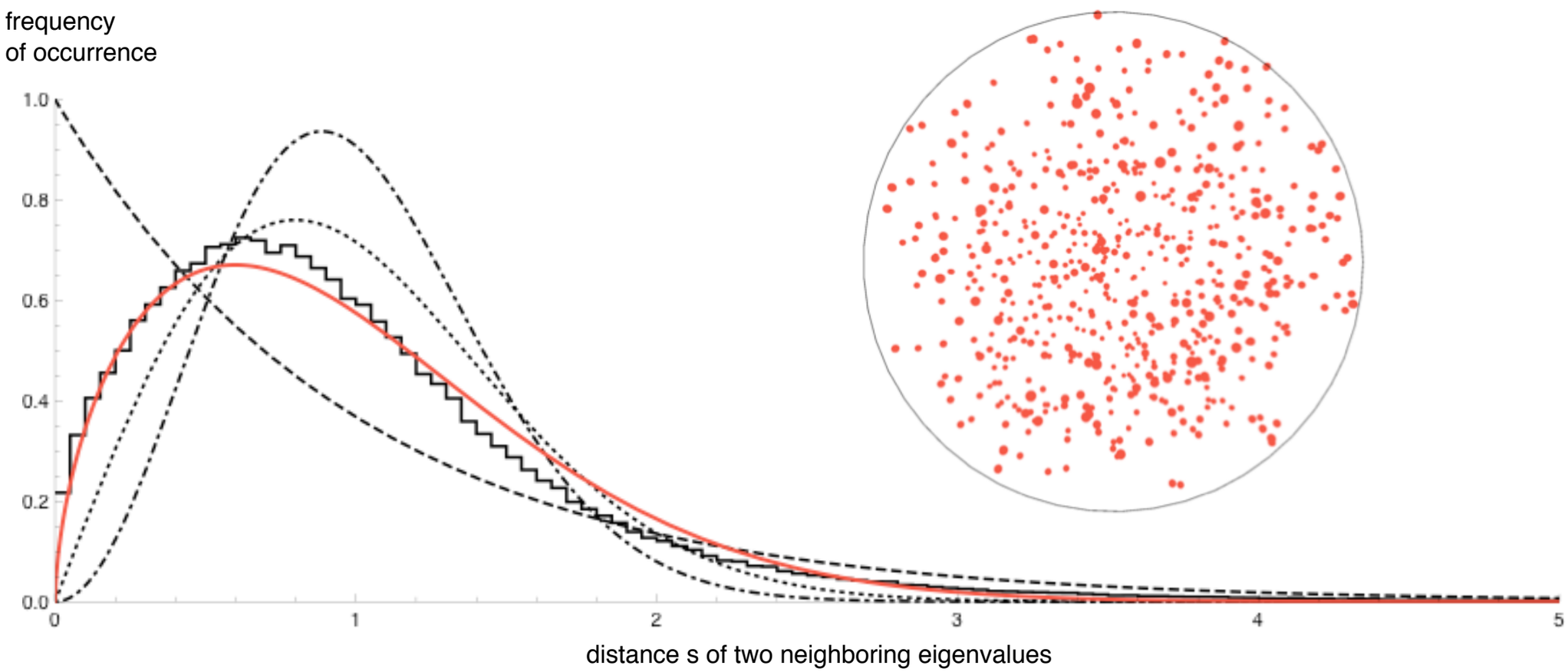
$$p(H_{ij}) \propto \mathrm{e}^{-A \sum_{kl} H_{kl}^2}$$

$$p(s) = \frac{\pi}{2} s \, \mathrm{e}^{-\pi s^2/4}$$

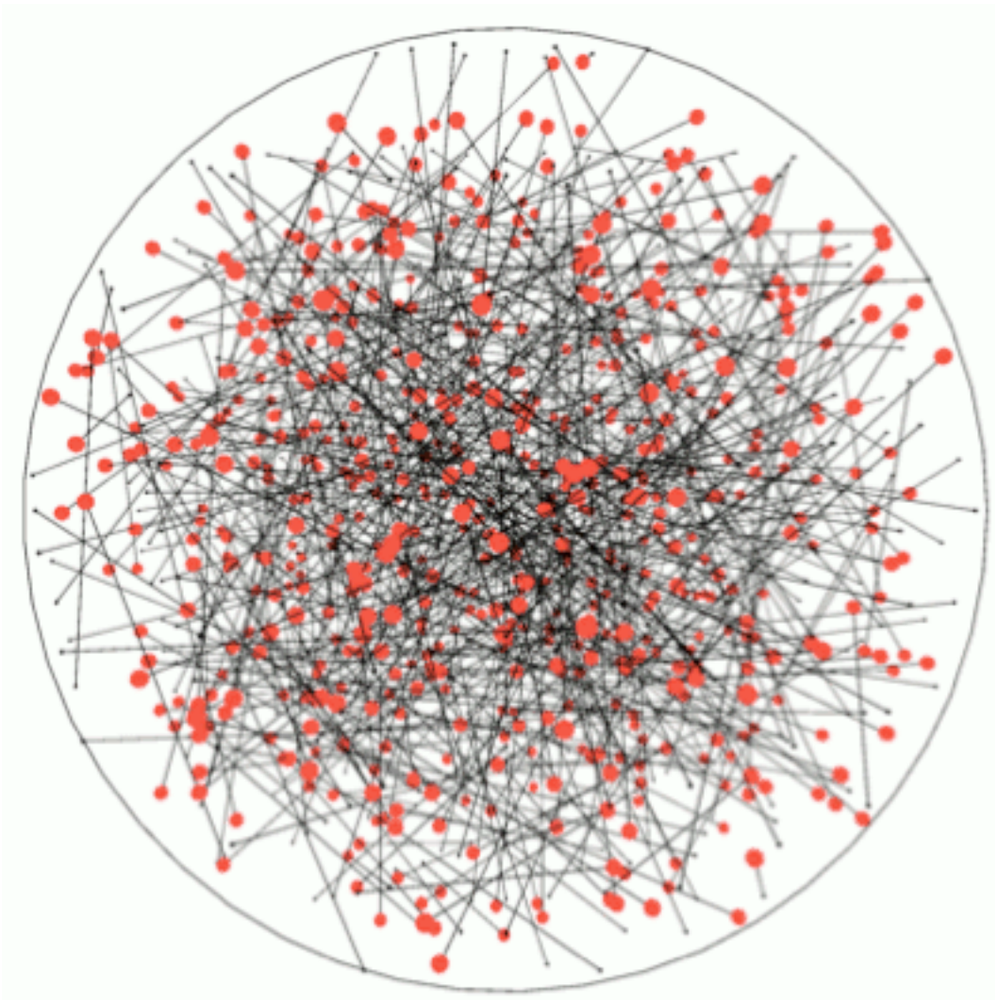
- Gaussian Unitary (GUE)

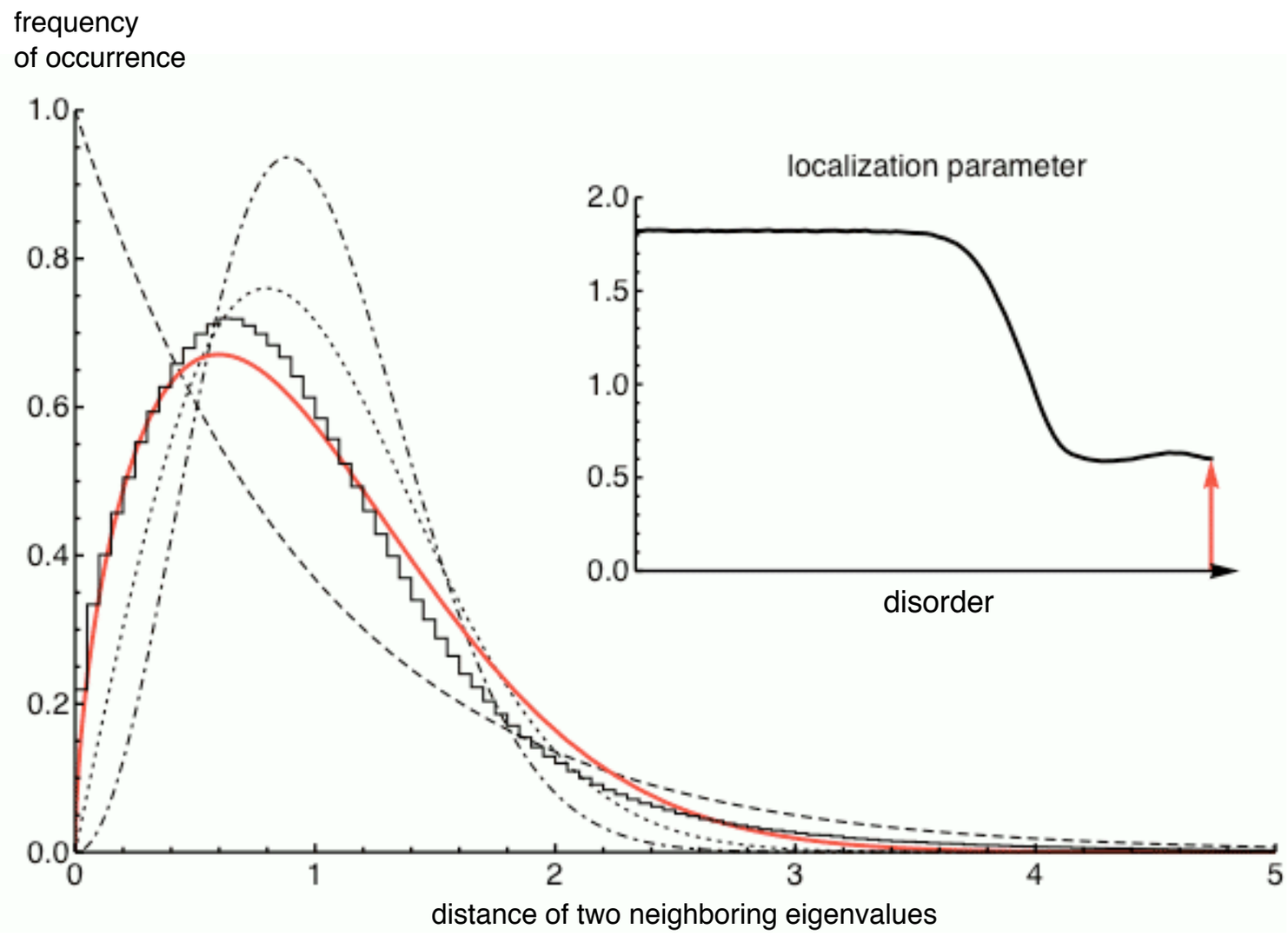
$$p(H_{ij}) \propto \mathrm{e}^{-A \sum_{kl} |H_{kl}|^2}$$

$$p(s) = \frac{32}{\pi^2} s^2 \, \mathrm{e}^{-4s^2/\pi}$$



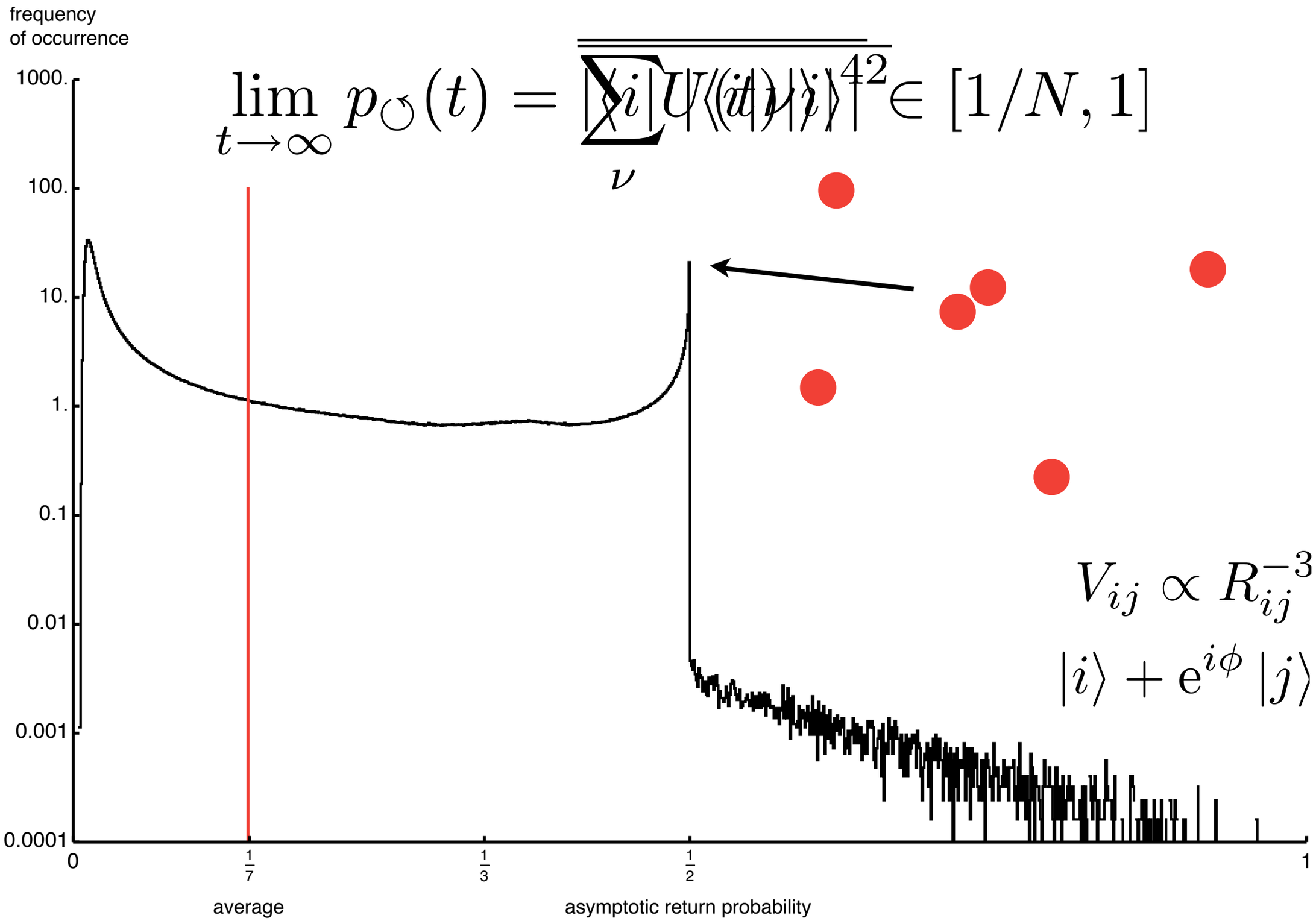
crossover




frequency
of occurrence
1.0
0.8
0.6
0.4
0.2
0.0
0
1
2
3
4
5
distance of two neighboring eigenvalues
localization parameter
2.0
1.5
1.0
0.5
0.0
disorder

return probability

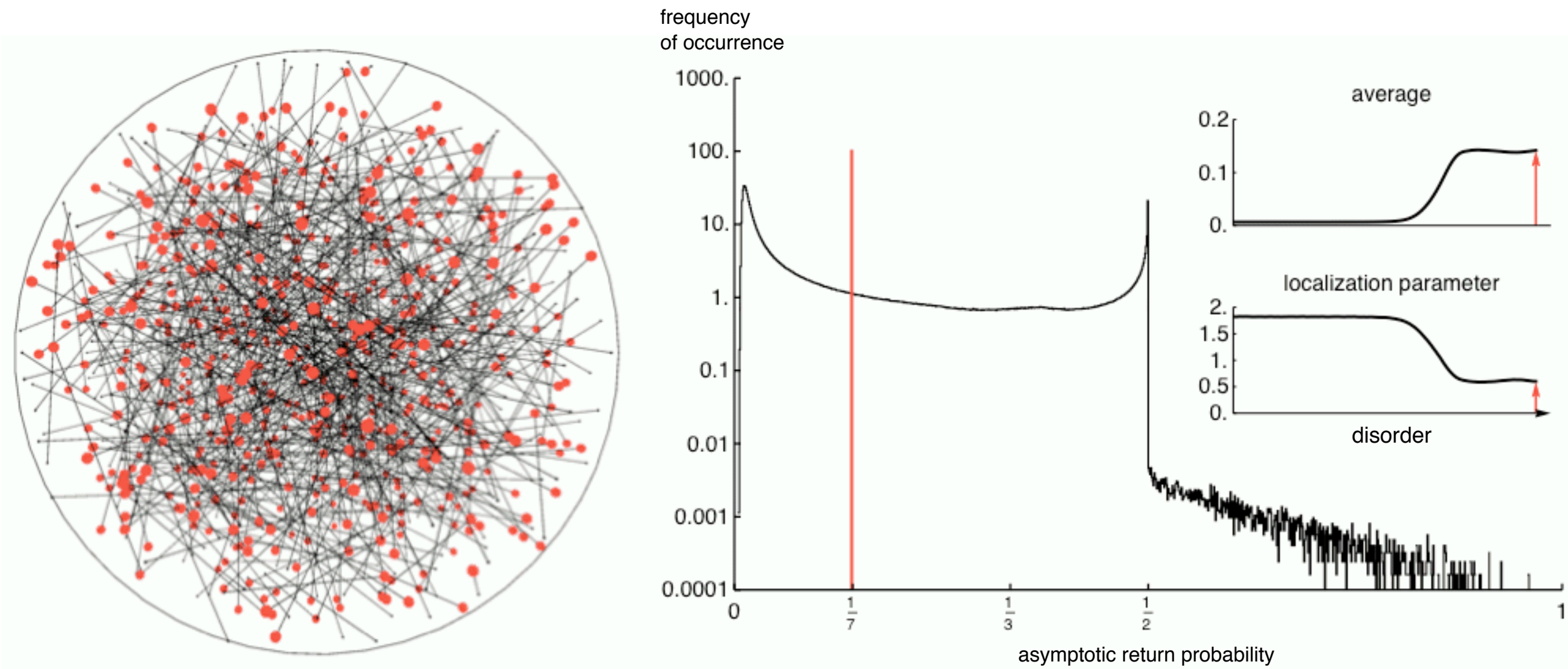
$$\lim_{t\to\infty} p_{\circlearrowleft}(t) = \overline{\overline{\sum_{\nu} |\langle i|U(t)|i\rangle|^2}} \in [1/N, 1]$$



$$V_{ij} \propto R_{ij}^{-3}$$

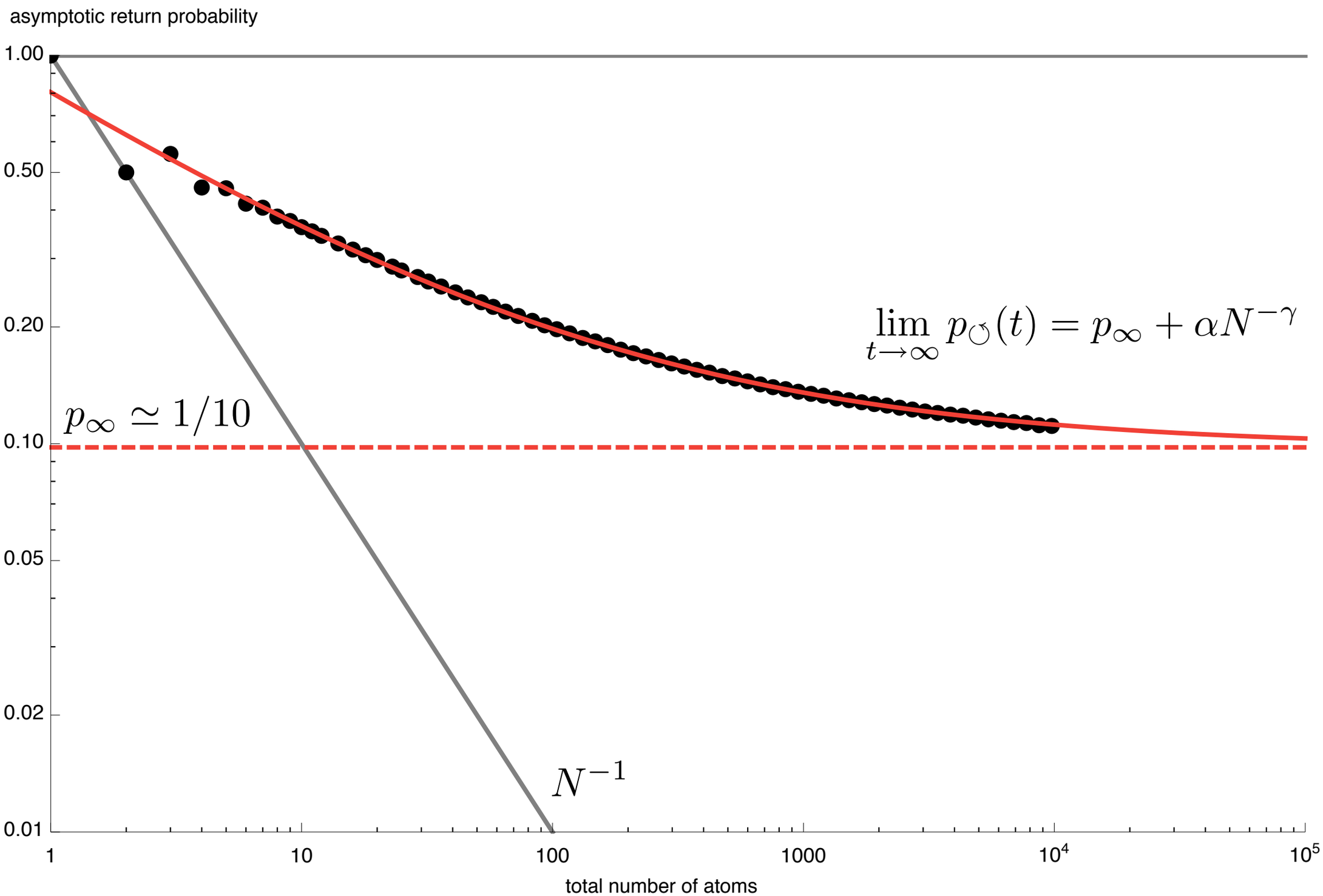
$$|i\rangle + \mathrm{e}^{i\phi}\,|j\rangle$$

crossover take 2



$$\lim_{t\to\infty} p_{\circlearrowleft}(t) = \overline{\sum_{\nu} |\langle i|\nu\rangle|^4}$$

one last thing...



abridged

- testing ground
- sensitivity to spatial disorder
- localization ↔ proximity of atoms
- experimental observation

Thanks for listening

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